

Experiment 17

Question

Perform Edge Detection using Sobel Matrix along X-axis.

Aim

To detect the vertical edges of an image using the Sobel operator along the X-axis.

Procedure

1. Import the OpenCV (cv2) library.
2. Read the input image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Read the image in grayscale mode.
5. Apply the Sobel operator along the X-axis.
6. Display the original and processed images.
7. Save the output image.

Code

```
import cv2

img = cv2.imread("dog.jpg", 0)

sobelx = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)

cv2.imshow("Original", img)
cv2.imshow("Sobel X", sobelx)

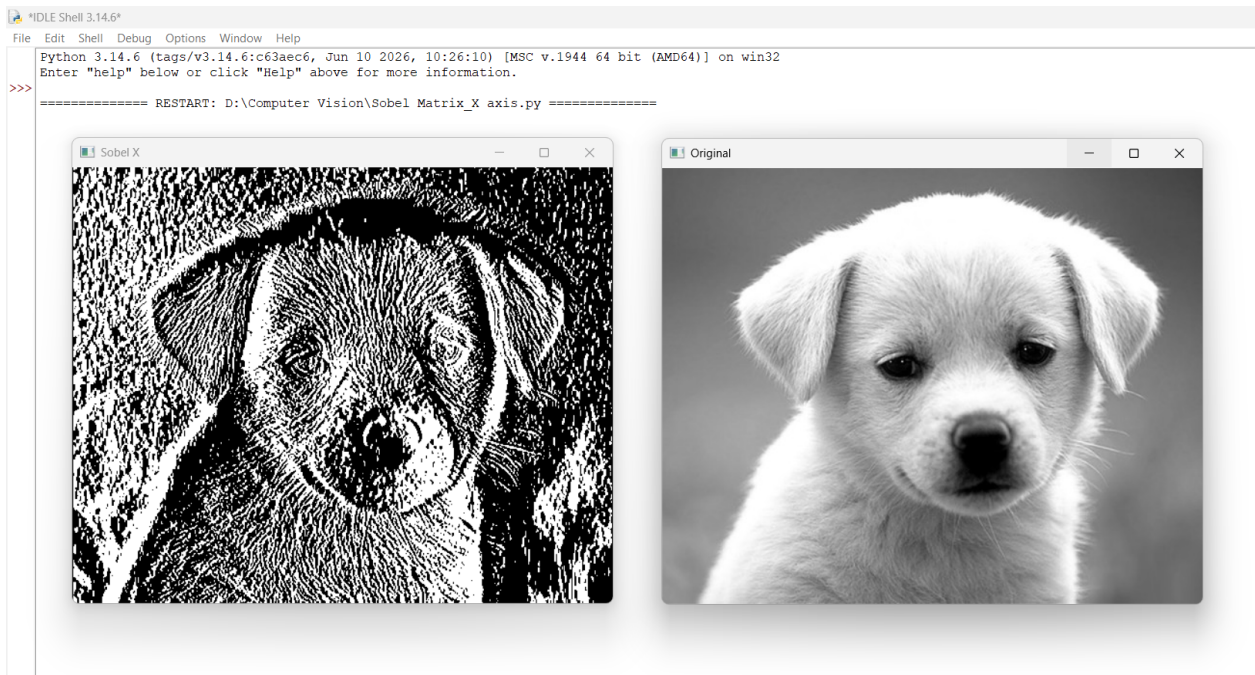
cv2.imwrite("sobelx_dog.jpg", sobelx)
```

Input

Input Image: dog.jpg

Output

- Sobel X Edge Image



Result

The vertical edges of the image were successfully detected using the Sobel operator along the X-axis.